

ADVERTISEMENT

famotidine (Rx, OTC)

Brand and Other Names: Pepcid AC, Zantac 360**Classes:** Histamine H2 Antagonists

Dosing & Uses

ADULT

Dosage Forms & Strengths

solution intravenous (Rx)

- 10mg/mL
- 4mg/mL
- 0.4mg/mL

oral suspension (Rx)

- 40mg/5mL (generic)

tablet (OTC)

- 10mg (Pepcid AC, Zantac 360, generic)
- 20mg (Pepcid AC, Zantac 360, generic)
- 40mg (Pepcid AC, generic)

Duodenal Ulcer

Acute treatment: 20 mg PO/IV q12hr or 40 mg PO at bedtime for 8 weeks

Maintenance: 20 mg PO at bedtime

Reduction of recurrence risk: 20 mg PO qDay for 1 year or as clinically indicated

Benign Gastric Ulcer

40 mg PO at bedtime up to 8 weeks

Gastroesophageal Reflux Disease

Nonerosive: 20 mg q12hr; up to 6 weeks

Erosive diagnosed by endoscopy: 20-40 mg PO q12hr for up to 12 weeks

Hypersecretory Conditions

20 mg PO/IV q6hr; may increase up to 160 mg q6hr

Heartburn

10-20 mg q12 hr; may take 15-60 min before eating foods that could cause heartburn

Dosing Modifications

CrCl <50 mL/min: Give 50% of usual dose, or prolong dosing interval to q36-48hr

PEDIATRIC

Dosage Forms & Strengths

solution intravenous

- 10mg/mL
- 4mg/mL
- 0.4mg/mL

oral suspension

- 40mg/5mL (generic)

tablet

- 10mg (Pepcid AC, Zantac 360, generic)
- 20mg (Pepcid AC, Zantac 360, generic)
- 40mg (Pepcid AC, generic)

Peptic Ulcer

1-17 years: 0.25 mg/kg IV q12hr or 0.5 mg/kg PO at bedtime; may increase to 1 mg/kg once daily at bedtime or 0.5 mg/kg twice daily for up to 8 weeks; not to exceed 40

mg/day

Gastroesophageal Reflux Disease

<3 months: 0.5 mg/kg PO once daily; may increase to 1 mg/kg once daily for up to 8 weeks

3-12 months: 0.5 mg/kg PO q12hr; may increase to 1 mg/kg twice daily; for up to 8 weeks; not to exceed 40 mg/day

1-17 years with or without esophagitis and ulceration: 0.5 mg/kg twice daily; for 6-12 weeks; not to exceed 40 mg twice daily

Heartburn

<12 years: Not established

>12 years: 10-20 mg q12 hr; may take 15-60 min before eating foods that could cause heartburn

Interactions

Interaction Checker

Enter a drug name to check for any interactions. and famotidine

No Interactions Found

Interactions Found

Contraindicated

Serious

Significant - Monitor Closely

Minor

All Interactions Sort By:

Contraindicated (0)

Serious (25)

- atazanavir
famotidine will decrease the level or effect of atazanavir by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Atazanavir solubility decreases as pH increases. Substantially reduced plasma concentrations of atazanavir are expected if H2-receptor antagonists (H2RA) are coadministered.

For treatment-naïve patients, take atazanavir simultaneously with the H2RA or at least 10 h afterwards. See dosage adjustment recommendations if coadministered in treatment-experienced patients.

- **bosutinib**
famotidine will decrease the level or effect of bosutinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **dapsone**
famotidine will decrease the level or effect of dapsone by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **dasatinib**
famotidine will decrease the level or effect of dasatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **defactinib**
famotidine decreases levels of defactinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Defactinib exposure may decrease when coadministered with gastric acid reducing agents, (eg, H2-blockers). If concomitant use cannot be avoided, separate administration by at least 2 hr.
- **digoxin**
famotidine will increase the level or effect of digoxin by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **infigratinib (DSC)**
famotidine will decrease the level or effect of infigratinib (DSC) by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. If use with an acid-reducing agent cannot be avoided, administer infigratinib 2 hr before or 10 hr after administration of a H2-antagonist.
- **itraconazole**
famotidine will decrease the level or effect of itraconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **ketoconazole**
famotidine will decrease the level or effect of ketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **levoketoconazole**
famotidine will decrease the level or effect of levoketoconazole by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **mefloquine**
mefloquine increases toxicity of famotidine by QTc interval. Avoid or Use Alternate Drug. Mefloquine may enhance the QTc prolonging effect of high risk QTc prolonging agents.
- **neratinib**
famotidine will decrease the level or effect of neratinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.

- pazopanib
famotidine will decrease the level or effect of pazopanib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Avoid coadministration of pazopanib with drugs that raise gastric pH; consider short-acting antacids in place of PPIs and H2 antagonists; separate antacid and pazopanib dosing by several hours
- pexidartinib
famotidine will increase the level or effect of pexidartinib by inhibition of GI absorption. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Separate pexidartinib by 2 hr before or 10 hr after taking an H2-antagonist.
- pimozone
famotidine, pimozone. Either increases toxicity of the other by QTc interval. Avoid or Use Alternate Drug.

pimozone, famotidine. Either increases toxicity of the other by QTc interval. Avoid or Use Alternate Drug.
- ponatinib
famotidine decreases levels of ponatinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- risedronate
famotidine will increase the level or effect of risedronate by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. Applies only to delayed release formulation; accelerates pH-sensitive dissolution of delayed release risedronate
- secretin
famotidine, secretin. Other (see comment). Avoid or Use Alternate Drug. Comment: Concomitant use of H2-receptor antagonists may cause a hyperresponse in gastrin secretion in response to stimulation testing with secretin, falsely suggesting gastrinoma. Discontinue H2-receptor antagonists at least 2 days before administering secretin to aid in the diagnosis of gastrinoma.
- sotorasib
famotidine will decrease the level or effect of sotorasib by affecting hepatic/intestinal enzyme CYP3A4 metabolism. Avoid or Use Alternate Drug. If use with an acid-reducing agent cannot be avoided, administer sotorasib 4 hr before or 10 hr after administration of a locally-acting antacid.
- sparsentan
famotidine decreases effects of sparsentan by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug. H2-antagonists may decrease sparsentan exposure which may reduce efficacy of sparsentan.
- tafenoquine
tafenoquine will increase the level or effect of famotidine by Other (see comment). Avoid or Use Alternate Drug. Tafenoquine inhibits organic cation transporter-2 (OCT2) and multidrug and toxin extrusion (MATE) transporters in vitro. Avoid coadministration with OCT2 or MATE substrates. If coadministration cannot be

avoided, monitor for substrate-related toxicities and consider dosage reduction if needed based on product labeling of the coadministered drug.

- **taletrectinib**
famotidine decreases levels of taletrectinib by increasing gastric pH. Applies only to oral form of both agents. Avoid or Use Alternate Drug.
- **trilaciclib**
trilaciclib will decrease the level or effect of famotidine by Other (see comment). Avoid or Use Alternate Drug. Avoid coadministration of trilaciclib (OCT2, MATE1, and MATE-2K inhibitor) with substrates where minimal increased concentration in kidney or blood may lead to serious or life-threatening toxicities.
- **vandetanib**
famotidine, vandetanib. Either increases toxicity of the other by QTc interval. Avoid or Use Alternate Drug.

vandetanib, famotidine. Either increases toxicity of the other by QTc interval. Avoid or Use Alternate Drug.
- **ziftomenib**
famotidine decreases levels of ziftomenib by unspecified interaction mechanism. Avoid or Use Alternate Drug. If unable to avoid, administer ziftomenib 2 hours before or 10 hours after administration of an H2 receptor antagonist.

Monitor Closely (44)

- **acalabrutinib**
famotidine decreases levels of acalabrutinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Acalabrutinib solubility decreases with increasing gastric pH. Administer acalabrutinib 2 hr before an H2-receptor antagonist.
- **budesonide**
famotidine decreases effects of budesonide by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Enteric-coated budesonide dissolves at pH >5.5. Also, dissolution of extended-release budesonide tablets is pH dependent. Coadministration with drugs that increase gastric pH may cause these budesonide products to prematurely dissolve, and possibly affect release properties and absorption of the drug in the duodenum.
- **carbonyl iron**
famotidine will decrease the level or effect of carbonyl iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- **cefdinir**
famotidine will decrease the level or effect of cefdinir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- **cefditoren**
famotidine will decrease the level or effect of cefditoren by increasing gastric pH.

Applies only to oral form of both agents. Use Caution/Monitor.

- cefpodoxime
famotidine will decrease the level or effect of cefpodoxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- cefuroxime
famotidine will decrease the level or effect of cefuroxime by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- crizotinib
famotidine decreases levels of crizotinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that elevate the gastric pH may decrease the solubility of crizotinib and subsequently reduce its bioavailability. However, no formal studies have been conducted. .
- cyclosporine
famotidine will increase the level or effect of cyclosporine by unknown mechanism. Use Caution/Monitor. Delayed resorption of cyclosporine has been reported when famotidine is coadministered with cyclosporine.
- dabrafenib
famotidine will decrease the level or effect of dabrafenib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Drugs that alter upper GI tract pH (eg, PPIs, H2-blockers, antacids) may decrease dabrafenib solubility and reduce its bioavailability
- dexamethylphenidate
famotidine will increase the level or effect of dexamethylphenidate by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Applies only to extended release formulation
- dordaviprone
dordaviprone and famotidine both increase QTc interval. Use Caution/Monitor. Increase the frequency of ECG and electrolyte monitoring.
- erdafitinib
famotidine increases levels of erdafitinib by decreasing renal clearance. Modify Therapy/Monitor Closely. Consider alternatives that are not OCT2 substrates or consider reducing the dose of OCT2 substrates based on tolerability.
- erlotinib
famotidine decreases levels of erlotinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Separate administration by taking erlotinib 2 hr before or 10 hr after H2 blockers. Coadministration of H2-blockers and erlotinib may decrease erlotinib exposure and reduce its efficacy.
- ferric maltol
famotidine will decrease the level or effect of ferric maltol by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous fumarate

famotidine will decrease the level or effect of ferrous fumarate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- ferrous gluconate
famotidine will decrease the level or effect of ferrous gluconate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ferrous sulfate
famotidine will decrease the level or effect of ferrous sulfate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- fosamprenavir
famotidine will decrease the level or effect of fosamprenavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- gefitinib
famotidine decreases levels of gefitinib by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor. Separate gefitinib and H2-antagonist doses by at least 6 hr.
- glipizide
famotidine will increase the level or effect of glipizide by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- glyburide
famotidine will increase the level or effect of glyburide by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- iron dextran complex
famotidine will decrease the level or effect of iron dextran complex by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- iron sucrose
famotidine will decrease the level or effect of iron sucrose by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- ledipasvir/sofosbuvir
famotidine decreases levels of ledipasvir/sofosbuvir by Other (see comment). Use Caution/Monitor. Comment: Ledipasvir solubility decreases as pH increases; drugs that increase gastric pH are expected to decrease levels of ledipasvir; H2-receptor antagonists may be administered simultaneously with or 12 hr apart from ledipasvir/sofosbuvir at a dose that does not exceed doses comparable to famotidine 40 mg BID.
- mesalamine
famotidine will decrease the level or effect of mesalamine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely.
- metformin
famotidine increases levels of metformin by Other (see comment). Use Caution/Monitor. Comment: Famotidine inhibits multidrug and toxin extrusion

(MATE) transporter 1, possibly increasing exposure to metformin. Monitor for metformin-related toxicities, including lactic acidosis.

- methylphenidate

famotidine will increase the level or effect of methylphenidate by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Applies only to extended release formulation

famotidine decreases effects of methylphenidate by enhancing GI absorption. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Since the characteristics of methylphenidate extended release capsules (Ritalin LA) are pH dependent, coadministration of antacids or acid suppressants could alter the release of methylphenidate. Consider separating the administration of the antacid and the methylphenidate extended-release capsules may be avoided.

- mifepristone

famotidine, mifepristone. Either increases toxicity of the other by QTc interval. Use Caution/Monitor.

mifepristone, famotidine. Either increases toxicity of the other by QTc interval. Use Caution/Monitor.

- mycophenolate

famotidine will decrease the level or effect of mycophenolate by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- nelfinavir

famotidine will decrease the level or effect of nelfinavir by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- nilotinib

famotidine decreases levels of nilotinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Avoid this interaction by administering H2 antagonists 10 hr after or 2 hr before nilotinib (ODT nilotinib tablets not affected by PPIs, H2-blockers, or antacids).

- polysaccharide iron

famotidine will decrease the level or effect of polysaccharide iron by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- posaconazole

famotidine will decrease the level or effect of posaconazole by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.

- pretomanid

pretomanid will increase the level or effect of famotidine by Other (see comment). Modify Therapy/Monitor Closely. In vitro studies demonstrated that pretomanid significantly inhibits OAT3; monitor for increased adverse effects and consider dosage reduction for OAT3 substrates.

- rilpivirine

famotidine will decrease the level or effect of rilpivirine by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely.

Concurrent use, may cause treatment failure and/or the development of rilpivirine or NNRTI resistance owing to decreased levels. Administer H2 antagonists at least 12 hours before or at least 4 hours after rilpivirine.

- rilzabrutinib
famotidine decreases levels of rilzabrutinib by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Give rilzabrutinib 2 hr before H2 blockers. Coadministration may decrease rilzabrutinib exposure, resulting in reduced efficacy.
- rose hips
famotidine will decrease the level or effect of rose hips by increasing gastric pH. Applies only to oral form of both agents. Use Caution/Monitor.
- serdexmethylphenidate/dexmethylphenidate
famotidine will increase the level or effect of serdexmethylphenidate/dexmethylphenidate by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Applies only to extended release formulation
- sofosbuvir/velpatasvir
famotidine will decrease the level or effect of sofosbuvir/velpatasvir by increasing gastric pH. Applies only to oral form of both agents. Modify Therapy/Monitor Closely. Velpatasvir solubility decreases as gastric pH increases (practically insoluble at pH >5). H2 receptor antagonists may be administered simultaneously with or 12 hr apart from sofosbuvir/velpatasvir at a dose that does not exceed doses comparable to famotidine 40 mg BID.
- taletrectinib
taletrectinib and famotidine both increase QTc interval. Use Caution/Monitor. Concomitant use may increase the risk of QTc interval prolongation. If concomitant use is unavoidable, adjust the frequency of ECG and electrolyte monitoring.
- vadadustat
vadadustat will increase the level or effect of famotidine by Other (see comment). Use Caution/Monitor. Vadadustat may increase exposure of OAT3 substrates. Monitor for signs of adverse effect of OAT3 substrate and reduce substrate dose in accordance with their product labeling.
- varenicline
famotidine will increase the level or effect of varenicline by decreasing renal clearance. Use Caution/Monitor.
- ziftomenib
famotidine, ziftomenib. QTc interval. Use Caution/Monitor. Caution if ziftomenib is coadministered with drugs that exacerbated risk of QT prolongation (eg, hypokalemia, orthostatic hypotension, inhibits metabolism of ziftomenib).

Minor (9)

- alendronate

famotidine increases levels of alendronate by unspecified interaction mechanism. Minor/Significance Unknown. Monitor for increase in alendronate side effects.

- aripiprazole
famotidine decreases levels of aripiprazole by unspecified interaction mechanism. Minor/Significance Unknown.
- axitinib
famotidine will decrease the level or effect of axitinib by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.
- blessed thistle
blessed thistle decreases effects of famotidine by pharmacodynamic antagonism. Minor/Significance Unknown. Theoretical interaction.
- ceftibuten
famotidine will decrease the level or effect of ceftibuten by increasing gastric pH. Applies only to oral form of both agents. Minor/Significance Unknown.
- cyanocobalamin
famotidine decreases levels of cyanocobalamin by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.
- devil's claw
devil's claw decreases effects of famotidine by pharmacodynamic antagonism. Minor/Significance Unknown.
- isavuconazonium sulfate
isavuconazonium sulfate will increase the level or effect of famotidine by Other (see comment). Minor/Significance Unknown. Isavuconazonium sulfate, an OCT2 inhibitor, may increase the effects or levels of OCT2 substrates.
- phytoestrogens
famotidine decreases levels of phytoestrogens by inhibition of GI absorption. Applies only to oral form of both agents. Minor/Significance Unknown.

Adverse Effects

1-10%

Headache (4.7%)

Diarrhea (1.7%)

Dizziness (1.3%)

Constipation (1.2%)

Frequency Not Defined

Body as a whole: Fever, asthenia, fatigue

Cardiovascular: Arrhythmia, AV block, palpitation; prolonged QT interval in patients with impaired renal function, has been reported very rarely

Gastrointestinal: Cholestatic jaundice, hepatitis, liver enzyme abnormalities, vomiting, nausea, abdominal discomfort, anorexia, dry mouth

Hematologic: Rare cases of agranulocytosis, pancytopenia, leukopenia, thrombocytopenia

Hypersensitivity: Anaphylaxis, angioedema, orbital or facial edema, urticaria, rash, conjunctival injection

Musculoskeletal: Rhabdomyolysis, musculoskeletal pain including muscle cramps, arthralgia

Nervous system/psychiatric: Grand mal seizure; psychic disturbances, which were reversible in cases for which follow-up was obtained, including hallucinations, confusion, agitation, depression, anxiety, decreased libido; paresthesia; insomnia; somnolence; convulsions, in patients with impaired renal function, have been reported very rarely

Respiratory: Bronchospasm, interstitial pneumonia

Skin: Toxic epidermal necrolysis/Stevens-Johnson syndrome (very rare), alopecia, acne, pruritus, dry skin, flushing

Special senses: Tinnitus, taste disorder

Rare cases of impotence and rare cases of gynecomastia

Warnings

Contraindications

Hypersensitivity to famotidine or other H₂-receptor antagonists

Cautions

Use caution in renal impairment; dosage adjustment recommended in moderate to severe renal impairment (CrCl <50 mL/min)

Prolonged QT interval reported rarely in patients with renal impairment whose dose or dosing interval may not have been adjusted appropriately

Central nervous system (CNS) adverse effects, including confusion, delirium, hallucinations, disorientation, agitation, seizures, and lethargy, reported with moderate-to-severe renal impairment; since famotidine blood levels are higher in patients with renal impairment than in patients with normal renal function, dosage adjustments are recommended in patients with renal impairment

Relief of symptoms does not eliminate the presence of gastric malignancy; consider evaluation for gastric malignancy in adult patients who have a suboptimal response or an early symptomatic relapse after completing treatment

State of confusion reported with use; risk increased in >50 years of age and/or renal/hepatic impairment

Prolonged treatment (>2 years) may lead to vitamin B12 malabsorption, which can result in vitamin B12 deficiency; magnitude of deficiency is dose related; occurs most frequently in females and those younger than 30 years

Patients should not use OTC if difficulty swallowing, vomiting blood, have bloody or black stools

If patient taking a prescription drug, the patient should ask a doctor or a pharmacist whether acid reducers can be taken concomitantly with it

Patients with kidney disease should ask doctor before use

Drug interaction overview

- Treatment can reduce absorption of other drugs, due to effect on reducing intragastric acidity, leading to loss of efficacy of the concomitant drug; concomitant administration with dasatinib, delavirdine mesylate, cefditoren, and fosamprenavir not recommended
- See prescribing information for other drugs dependent on gastric pH for absorption for administration instructions, including atazanavir, erlotinib, ketoconazole, itraconazole, ledipasvir/sofosbuvir, nilotinib, and rilpivirine
- Although not studied clinically, drug is considered a weak CYP1A2 inhibitor and may lead to substantial increases in blood concentrations of tizanidine, a CYP1A2 substrate; avoid concomitant use; if concomitant use necessary, monitor for hypotension, bradycardia or excessive drowsiness; refer to full prescribing information for tizanidine

Pregnancy & Lactation

Pregnancy

Available data in pregnant women are insufficient to establish a drug-associated risk of major birth defects, miscarriage or adverse maternal or fetal outcomes

Animal data

- Animal reproduction studies have shown no adverse development effects at doses up to approximately 243 times, the recommended human dose of 80 mg per day for treatment of erosive esophagitis

Lactation

There are limited data available on presence in human breast milk; there were no effects on breastfed infant; there are no data on famotidine effects on milk production; drug reported present in milk of lactating rats; developmental and health benefits of breastfeeding should be considered along with mother's clinical need for famotidine and any potential adverse effects on breastfed child or from underlying maternal condition

Pregnancy Categories

A: Generally acceptable. Controlled studies in pregnant women show no evidence of fetal risk.

B: May be acceptable. Either animal studies show no risk but human studies not available or animal studies showed minor risks and human studies done and showed no risk.

C: Use with caution if benefits outweigh risks. Animal studies show risk and human studies not available or neither animal nor human studies done.

D: Use in LIFE-THREATENING emergencies when no safer drug available. Positive evidence of human fetal risk.

X: Do not use in pregnancy. Risks involved outweigh potential benefits. Safer alternatives exist.

NA: Information not available.

Pharmacology

Mechanism of Action

Blocks H₂ receptors of gastric parietal cells, leading to inhibition of gastric secretions

Absorption

Bioavailability: 40-45% (PO; minimal 1st-pass metabolism)

Onset: <1 hr (PO); <30 min (IV)

Duration: 10-12 hr

Peak plasma time: IV, 20 min; PO, 1-4 hr

Distribution

Protein bound: 15-20%

V_d: 1.1-1.4 L/kg (Adults); 1.5-2.07 L/kg (children); 1.4-1.8 L/kg (infants <3 months); 2.3 L/kg (infants 3-12 months)

Metabolism

Metabolized in liver

Metabolites: Famotidine S-oxide (inactive)

Elimination

Half-life: 2.5-4 hr (adults; increases with renal impairment; eg, 20 hr with CrCl <10 mL/min); 3-4 hr (children); 4.5 hr (infants 3-12 months); 8-10.5 hr (infants < 3 months)

Dialyzable: No

Renal clearance: 250-450 mL/min

Total body clearance: 381-483 mL/min

Excretion: Urine (25-30% as unchanged drug when administered PO; 70% when administered IV)

Administration

IV Incompatibilities

Additive: Piperacillin-tazobactam

Y-site: Cefepime, piperacillin-tazobactam, amphotericin B, azithromycin, furosemide (at 2 mg/mL famotidine; compatible at 0.2 mg/mL)

IV Compatibilities

Additive: Aztreonam, ceftazidime, dobutamine, dopamine, furosemide, gentamicin, imipenem, thiamine

Y-site: Atropine, cefazolin, furosemide, gentamicin, heparin, norepinephrine, thiamine

IV Preparation

Dilute 20 mg to total of 5 or 10 mL with NS, D5W, or LR

Also available in premixed bag containing 20 mg in 50 mL NS

IV Administration

Infuse at rate no faster than 10 mg/min

Storage

Premixed: Store at room temperature

Unmixed: Store in refrigerator at 2-8°C (36-46°F)

Formulary

Adding plans allows you to compare formulary status to other drugs in the same class.

Formulary information is available for health plans only in the United States.

Medscape prescription drug monographs are based on FDA-approved labeling information, unless otherwise noted, combined with additional data derived from primary medical literature.
